Ecological Niche Modeling of *Vespa velutina*: Methodology, Theory, and Research Assets

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Introducción

Limpieza de datos

Modelo

Why not deep learning? (yet)

Feature engineering

Limpieza general

Pseudo-absences

Model A (Barbet-Massin)

To effectively model the potential niche of the *Vespa velutina*, the algorithm must distinguish between suitable and unsuitable habitats. This requires both presence and absence data. However, obtaining "true absences" (ground truth) is logistically challenging and expensive, because not sighting a hornet does not confirm its absence. Therefore, the standard practice is to generate pseudo-absences: synthetic points where the species is assumed to be absent.

While "presence-only" models like Maxent exist, literature consistently shows that presence-absence frameworks yield superior predictive performance. Nevertheless, generating these points is not a trivial task; simply scattering them randomly often fails to capture the species' complex invasive dynamics. To avoid arbitrary decisions, this project follows the guidelines established by the British Royal Society in the seminal paper: *'Selecting pseudo-absences for species distribution models: how, where and how many?'*.

A key takeaway from this research is that the sample size of pseudo-absences often outweighs the specific generation technique in terms of model impact. Furthermore, instead of using a single large pool of absences, the study suggests training in "batches" and averaging the results to minimize stochastic noise and improve accuracy. Since this project utilizes a Random Forest classifier, maintaining a 1:1 ratio (prevalence) is paramount. This balance, combined with a geographically and environmentally stratified selection, ensures the model learns the true biological limits of the species.

The following points outline the technical justification for each methodological choice based on the aforementioned study:

What is the optimal presence-to-absence ratio for Random Forest?

For Machine Learning algorithms like Random Forest, the optimal ratio is 1:1. According to Barbet-Massin et al., model accuracy (TSS) drops significantly when prevalence is low. Therefore, I have configured the dataset so the number of pseudo-absences matches the number of recorded nests (presences).

In fact, the study concludes that the quantity of pseudo-absences is, by far, the most influential factor in enhancing model precision for this algorithm.

How many times should pseudo-absence selection be replicated?

A single run is rarely optimal if the goal is to maximize model quality. To prevent random selection from "polluting" the results, I have implemented 5 replicates. The final map is an ensemble average of these runs, which enhances statistical robustness and filters out biases from random outliers.

The required number of replicates typically decreases as the number of pseudo-absences per batch increases. Given that our dataset is large (over 25,000 points) and we are using a 1:1 ratio, a small number of replicates is sufficient to reach a stable asymptote. This parameter is highly "fine-tunable," allowing for experimentation with 2, 3, or 5 replicates to determine which configuration yields the most consistent results.

What are the optimal number and weighting scheme for pseudo-absences per replicate?

For our Random Forest model, maximum accuracy is achieved when the number of pseudo-absences is approximately equal to the available presences. Given our robust dataset of ~25,000 observations, each replicate will contain exactly 25,000 pseudo-absences.

The study by Barbet-Massin et al. (2012) demonstrates that, unlike statistical models such as GLMs (which tolerate ratios of 1:100 or higher), Machine Learning algorithms like Random Forest suffer a drastic drop in sensitivity when an excessive number of absences is used. This occurs because the algorithm tends to overfit the majority class (absences), becoming unable to correctly identify high-risk areas.

Regarding weighting, the results indicate that the best performance is obtained by applying equal weights to both presences and pseudo-absences. Therefore, our ensemble strategy will average multiple replicates, each maintaining this critical 1:1 balance to ensure the Random Forest accurately learns the boundaries of the hornet's niche.

How do we choose where to place pseudo-absences?

For a Random Forest trained with a large number of presence points, the most effective method is Climate Exclusion (SRE - Surface Range Envelope). Selecting absences in areas that are environmentally distinct from presence locations significantly improves model accuracy compared to simple random sampling, which often risks generating "false absences" in potential expansion zones.

If we had fewer presence points, Geographic Exclusion (placing pseudo-absences at a specific spatial distance from known nests) would be more effective. This geographic approach could also be used to create a more "alarmist" model, prioritizing a higher rate of false positives over conservative estimates. However, given our high data volume, SRE ensures the model captures the true environmental niche.

What if my presence data is biased?

The Barbet-Massin study suggests that the number and weighting of pseudo-absences are largely unaffected by presence bias. However, if a significant climatic bias were present, geographic

exclusion might be preferable over climatic exclusion; this prevents the model from assuming the species can only survive in the specific climates where it was sighted.

In our case, while sightings are heavily concentrated in temperate Atlantic climates, this is unlikely to be a sampling bias. *Vespa velutina* clearly favors high humidity, dense vegetation, and temperate conditions over the arid or extreme cold environments found in the Castilian Plateau, the Alps, or the Mediterranean coast.

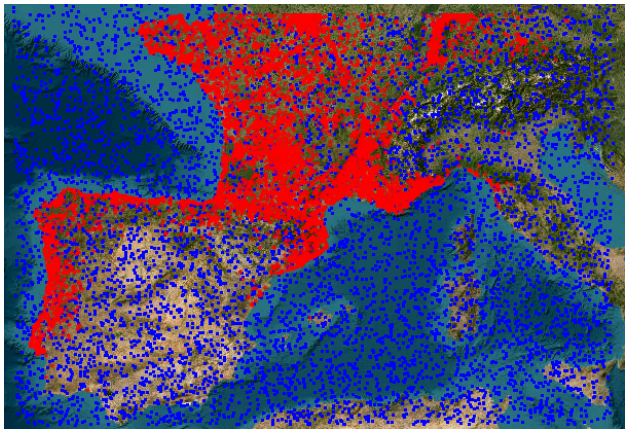
With over 25,000 data points and a decade of invasion history, the absence of the hornet in these regions is not due to a lack of observers—it is due to a direct physiological barrier. While this factor has a smaller impact on overall precision than the total count of pseudo-absences, accounting for it ensures our model reflects biological reality rather than human observation patterns.

Conclusion

Ultimately, to maximize the predictive power of our Random Forest, we have implemented an ensemble strategy based on a strict 1:1 class balance and an equal weighting scheme. By combining the robustness of 25,000 records with a Climate Exclusion (SRE) method and averaging multiple replicates, we ensure a high-sensitivity model that respects the actual physiological limits of *Vespa velutina*. In the end, averaging several model runs is not just about statistical safety—it is what makes the final map realistic, ensuring it doesn't depend on the luck of where a few random points happened to fall.

Modelo B (Barbet-Massin)

Another approach to generate pseudo-absences in invasive species



Librerías

Papers útiles (Buffer con info aplicable a todos los demás apartados)

Niche modelling and landscape genetics of the yellow-legged hornet (*Vespa velutina*): An integrative approach for evaluating central–marginal population dynamics in Europe:

El paper nos dice que la variación genética es algo fundamental para predecir cómo de bien se va a expandir una especie, y que esta variación se ve muy muy disminuída en los bordes

de la zona invadida por Vespa velutina. Los bordes de la expansión son genéticamente mucho menos diversas, porque menos individuos son capaces de adaptarse a ella. La diferencia ambiental también es fundamental para ver cómo evoluciona esta diversidad, como veremos a continuación.

¿Qué ocurre? Las especies invasoras son malas (expandir en esto),
Hay varios factores por los que una especie puede prosperar como invasora en un entorno extranjero, mientras que otros caen en desgracia.

Selecting pseudo-absences for species distribution models: how, where and how many?